

# Lecture 15: Conditioning of $Ax = b$ ; Stability of LU Factorization

Math 6610

C. Mou

Department of Mathematics and Statistics  
Utah State University

Fall 2026

# Today's Goals

- Review Lecture 14; which norm to use when
- Derive  $\kappa(A) = \|A\| \|A^{-1}\|$  for the problem  $Ax = b$
- Show  $\kappa(A) \geq 1$  and interpret extreme cases
- Explain precisely why LU without pivoting is not backward stable
- Define the growth factor  $\rho(A)$  and state the pivoted-LU stability theorem

# Review of Lecture 14

**Remark (Accuracy theorem).** A backward stable algorithm returns the exact answer for an input wrong by at most  $C \epsilon_{\text{mach}}$ ; the problem magnifies that by about  $\kappa(x)$ :

$$\underbrace{\frac{\|\tilde{F}(x) - F(x)\|}{\|F(x)\|}}_{\text{output error}} \leq 2\kappa(x) \cdot C \epsilon_{\text{mach}} \quad \text{for small } \epsilon_{\text{mach}}.$$

**Remark (lub norms).**  $\|A\| = \text{lub}(A) = \max_{\|x\|=1} \|Ax\|$  is the largest stretch of  $A$ . Rules used today:

$$\|Ax\| \leq \|A\| \|x\|, \quad \|AB\| \leq \|A\| \|B\|, \quad \|I\| = 1.$$

**Remark (The classical three).**  $\|A\|_1 = \max \text{column sum}$ ,  $\|A\|_\infty = \max \text{row sum}$ ,  $\|A\|_2 = \sigma_{\max}(A)$  (largest Euclidean stretch).

# Which Norm When?

**Proposition.** For all  $x \in \mathbb{C}^n$ :

$$\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1 \leq n \|x\|_\infty.$$

*Proof. 1. Largest entry  $\leq$  length.* Let  $|x_m| = \|x\|_\infty$ . One term  $\leq$  the sum; then take square roots:

$$\|x\|_\infty^2 = |x_m|^2 \leq \sum_i |x_i|^2 = \|x\|_2^2.$$

*2. Length  $\leq$  total size.* Cross terms are  $\geq 0$ ; then take square roots:

$$\|x\|_1^2 = \sum_i |x_i|^2 + \sum_{i \neq l} |x_i| |x_l| \geq \|x\|_2^2.$$

*3. Total size  $\leq n \times$  largest entry.* Each of the  $n$  terms is at most  $\|x\|_\infty$ , so  $\|x\|_1 = \sum_{i=1}^n |x_i| \leq n \|x\|_\infty$ . ■

# Which Norm When? (Cont'd)

**Remark** (*All norms are comparable; proof omitted*). For any two norms  $\|\cdot\|_a$ ,  $\|\cdot\|_b$  on  $\mathbb{C}^n$  there are  $c, c' > 0$  with  $c \|x\|_a \leq \|x\|_b \leq c' \|x\|_a$  for all  $x$ . So “ $= O(\epsilon_{\text{mach}})$ ” or “converges” in one norm means the same in every norm.

- **Constants change.**  $\kappa_1(A)$ ,  $\kappa_2(A)$ ,  $\kappa_\infty(A)$  differ in general, by factors that depend only on  $n$ . Say which norm you use.
- $\|\cdot\|_1$ ,  $\|\cdot\|_\infty$ : easy to compute (about  $n^2$  operations); good for bounds and in software.
- $\|\cdot\|_2$ : unchanged by unitary factors,  $\|UAV\|_2 = \|A\|_2$  for unitary  $U, V$  (rotations and reflections do not change it). The natural norm for eigenvalues, later in the course.
- $\|\cdot\|_F$ : also unchanged by unitary factors and easy to compute, but for  $n \geq 2$  not an lub norm of any vector norm:  $\|I\|_F = \sqrt{n} \neq 1 = \text{lub}(I)$ .

# Conditioning of $Ax = b$

**Remark (The problem).**  $A \in \mathbb{C}^{n \times n}$  nonsingular (invertible),  $b \in \mathbb{C}^n$  fixed with  $b \neq 0$ , and  $x := A^{-1}b$  (so  $x \neq 0$ ). Change  $A$  only:

$$(A + \delta A)(x + \delta x) = b.$$

$\|\cdot\|$ : any vector norm on  $\mathbb{C}^n$ ; on matrices, its lub norm.

**Definition (Condition number of a matrix).**

$$\kappa(A) := \|A\| \|A^{-1}\| \quad (\text{also written } \kappa_{\|\cdot\|}(A): \text{it depends on the norm}).$$

**Remark (Question).** How much can a small relative change in  $A$  move  $x$ ?

# Conditioning of $Ax = b$ : Intuition

**Remark (Intuition).** Subtract  $Ax = b$  from  $(A + \delta A)(x + \delta x) = b$ :

$$A \delta x + (\delta A) x + (\delta A)(\delta x) = 0.$$

$(\delta A)(\delta x)$  is (small)  $\times$  (small): drop it, multiply on the left by  $A^{-1}$ , take norms:

$$\delta x \approx -A^{-1}(\delta A) x \implies \frac{\|\delta x\|}{\|x\|} \lesssim \|A^{-1}\| \|\delta A\| = \kappa(A) \frac{\|\delta A\|}{\|A\|}.$$

Relative change in  $x \lesssim \kappa(A) \times$  relative change in  $A$ .

**Remark (Next).** The theorem makes “ $\lesssim$ ” exact and shows  $\kappa(A)$  cannot be replaced by anything smaller.

# Conditioning of $Ax = b$ : Theorem

**Theorem** (*Condition number of  $Ax = b$* ). With  $b$  fixed, the condition number of  $A \mapsto x = A^{-1}b$  (Lecture 9: input  $A$ , output  $x$ ) is  $\kappa(A)$ :

$$\lim_{\delta \rightarrow 0} \underbrace{\sup_{0 < \|\delta A\| \leq \delta} \frac{\|\delta x\| / \|x\|}{\|\delta A\| / \|A\|}}_{\text{worst amplification}} = \|A\| \|A^{-1}\| = \kappa(A).$$

(For  $\delta < 1/\|A^{-1}\|$ ,  $A + \delta A$  is invertible by Step 1, so  $\delta x$  exists.)

**Lemma** (*Norming vector; proof omitted*). For every  $x \neq 0$  there is  $w \in \mathbb{C}^n$  with

$$w^* x = \|x\| \quad \text{and} \quad |w^* z| \leq \|z\| \quad \text{for all } z \in \mathbb{C}^n$$

( $w^*$  = conjugate transpose). For  $\|\cdot\|_2$ :  $w = x / \|x\|_2$  (Cauchy–Schwarz).

# Conditioning of $Ax = b$ : Proof

*Proof. Idea:* upper bound from an exact formula for  $\delta x$ ; lower bound from one bad  $\delta A$ .

1. *Small changes keep  $A$  invertible.* Let  $t := \|A^{-1}\| \|\delta A\| < 1$ . If  $(A + \delta A)z = 0$ , then  $z = -A^{-1}(\delta A)z$ , so

$$\|z\| \leq t \|z\| \implies z = 0.$$

2. *Exact formula.* Subtract  $Ax = b$ , multiply on the left by  $A^{-1}$ :

$$\delta x = -A^{-1}(\delta A)(x + \delta x).$$

3. *Upper bound.* Take norms:  $\|\delta x\| \leq t (\|x\| + \|\delta x\|)$ . Solve for  $\|\delta x\|$  and divide by  $\|\delta A\| / \|A\|$ :

$$\frac{\|\delta x\| / \|x\|}{\|\delta A\| / \|A\|} \leq \frac{\kappa(A)}{1 - \|A^{-1}\| \|\delta A\|} \xrightarrow{\|\delta A\| \rightarrow 0} \kappa(A).$$

*(continued on the next slide)*

# Conditioning of $Ax = b$ : Proof (Cont'd)

*Proof (continued).* 4. *One bad change.* Pick  $y$  with  $\|y\| = 1$  and  $\|A^{-1}y\| = \|A^{-1}\|$  (the direction  $A^{-1}$  stretches most), and  $w$  from the lemma. For small  $\epsilon > 0$  let  $\delta A := -\epsilon y w^*$ :

$$\|\delta A\| = \epsilon, \quad (\delta A)x = -\epsilon \|x\| y.$$

(Check:  $\|\delta A z\| = \epsilon |w^* z| \leq \epsilon \|z\|$ , with equality at  $z = x$ .)

5. *Lower bound.* Step 2 reads  $\delta x = \epsilon \|x\| A^{-1}y - A^{-1}(\delta A) \delta x$ . Use

$\|u - v\| \geq \|u\| - \|v\|$ :

$$\|\delta x\| \geq \epsilon \|A^{-1}\| (\|x\| - \|\delta x\|) \implies \frac{\|\delta x\| / \|x\|}{\|\delta A\| / \|A\|} \geq \frac{\kappa(A)}{1 + \epsilon \|A^{-1}\|} \xrightarrow{\epsilon \rightarrow 0} \kappa(A).$$

6. *Squeeze.* With  $\epsilon = \delta$ : the worst amplification lies between the two bounds, and both tend to  $\kappa(A)$ . ■

$$\kappa(A) \geq 1$$

**Corollary.** For every nonsingular  $A \in \mathbb{C}^{n \times n}$  and every lub norm,  $\kappa(A) \geq 1$ .

*Proof.*  $\|I\| = 1$  and  $\|MN\| \leq \|M\| \|N\|$  (Lecture 14):

$$1 = \|I\| = \|AA^{-1}\| \leq \|A\| \|A^{-1}\| = \kappa(A).$$



**Remark** (*Extreme cases*).

- $\kappa(A) = 1$  (e.g.  $A = I$ ): relative changes in  $A$  are not magnified at all.
- $\kappa(A)$  large: about  $\log_{10} \kappa(A)$  digits of  $x$  are lost (Lecture 9). This happens when  $A$  is close to a singular matrix (made precise below).

# LU Without Pivoting: Backward Error

**Theorem (Proof omitted).** Let  $A \in \mathbb{C}^{n \times n}$  be nonsingular with  $A = LU$  ( $L$  lower triangular with ones on the diagonal,  $U$  upper triangular; i.e. elimination without row swaps meets no zero pivot). For all small enough  $\epsilon_{\text{mach}}$ , LU without pivoting in floating point runs to the end and returns  $\tilde{L}, \tilde{U}$  with

$$\tilde{L}\tilde{U} = A + \delta A, \quad \frac{\|\delta A\|}{\|L\| \|U\|} = O(\epsilon_{\text{mach}}).$$

**Remark (Meaning of  $O(\epsilon_{\text{mach}})$ ).**  $\|\delta A\| \leq C \epsilon_{\text{mach}} \|L\| \|U\|$  for small  $\epsilon_{\text{mach}}$ , with  $C$  depending on  $n$  but not on  $A$ .

**Remark (When is this good?).** If  $\|L\| \|U\| \leq c \|A\|$  with  $c$  independent of  $A$ , then  $\|\delta A\| / \|A\| \leq Cc \epsilon_{\text{mach}}$ : backward stable. Without pivoting no such  $c$  exists, and the example shows  $\delta A$  really can be large.

# LU Without Pivoting Fails: Example

**Example** (Double precision,  $\eta := 2^{-66} \approx 1.4 \cdot 10^{-20}$ ).

$$A = \begin{bmatrix} \eta & 1 \\ 1 & 1 \end{bmatrix}, \quad L = \begin{bmatrix} 1 & 0 \\ \eta^{-1} & 1 \end{bmatrix}, \quad U = \begin{bmatrix} \eta & 1 \\ 0 & 1 - \eta^{-1} \end{bmatrix}.$$

The multiplier  $\eta^{-1} = 2^{66}$  is exact. Near  $2^{66}$  doubles are  $2^{13}$  apart, so  $1 - 2^{66}$  rounds to  $-2^{66}$ :

$$\tilde{U} = \begin{bmatrix} \eta & 1 \\ 0 & -2^{66} \end{bmatrix}, \quad \tilde{L}\tilde{U} = \begin{bmatrix} \eta & 1 \\ 1 & 0 \end{bmatrix}, \quad \delta A = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}.$$

(Entry (2, 2):  $2^{66} \cdot 1 + (-2^{66}) = 0$ .) In the  $\infty$ -norm (max row sum),

$$\frac{\|\delta A\|_{\infty}}{\|A\|_{\infty}} = \frac{1}{2}, \quad \|L\|_{\infty} = 2^{66} + 1, \quad \|U\|_{\infty} = 2^{66} - 1.$$

# LU Without Pivoting Fails: Example (Cont'd)

**Example (Solving with  $b = (1, 2)^T$ ).** Exact:  $x = \left(\frac{1}{1-\eta}, \frac{1-2\eta}{1-\eta}\right)^T \approx (1, 1)^T$ .

Computed with  $\tilde{L}$ ,  $\tilde{U}$ :

$$\tilde{y} = \begin{bmatrix} 1 \\ \text{fl}(2 - 2^{66}) \end{bmatrix} = \begin{bmatrix} 1 \\ -2^{66} \end{bmatrix}, \quad \tilde{x}_2 = 1, \quad \tilde{x}_1 = \frac{1-1}{\eta} = 0.$$

So  $\tilde{x} = (0, 1)^T$ : no correct digit in  $\tilde{x}_1$ . Yet  $\kappa_\infty(A) = 2 \cdot \frac{2}{1-\eta} \approx 4$  (check:  $A^{-1} = \frac{1}{\eta-1} \begin{bmatrix} 1 & -1 \\ -1 & \eta \end{bmatrix}$ ).

**Remark (Not backward stable, not even stable).** A backward stable method would give error  $\leq 2 \cdot 4 \cdot C \epsilon_{\text{mach}}$  (accuracy theorem). And every matrix within relative distance  $O(\epsilon_{\text{mach}})$  of  $A$  has its solution within  $O(\epsilon_{\text{mach}})$  of  $(1, 1)^T$  (Step 3 of the proof), far from  $\tilde{x}$ : not stable either.

# Partial Pivoting and the Growth Factor

**Definition (Partial pivoting).** At elimination step  $k$ , swap row  $k$  with a row  $i \geq k$  whose entry in column  $k$  is largest in absolute value. Result:  $PA = LU$ , with  $P$  a permutation matrix (the identity with its rows reordered).

**Remark (Effect on  $L$ ).** Each multiplier is (entry)/(pivot), and the pivot is the largest entry:

$$|l_{jk}| \leq 1 \quad \text{for all } j, k \quad \implies \quad \|L\|_{\infty} \leq n.$$

What about  $U$ ?

**Definition (Growth factor).** With  $PA = LU$  from partial pivoting in exact arithmetic,

$$\rho = \rho(A) := \frac{\max_{j,k} |u_{jk}|}{\max_{j,k} |a_{jk}|} :$$

how much the entries grow during elimination.

# LU With Partial Pivoting: Backward Error

**Theorem (Proof omitted).** Let  $A \in \mathbb{C}^{n \times n}$  be nonsingular. LU with partial pivoting in floating point returns  $\tilde{L}$ ,  $\tilde{U}$  and a permutation  $\tilde{P}$  with

$$\tilde{L}\tilde{U} = \tilde{P}A + \delta A, \quad \frac{\|\delta A\|}{\|A\|} = O(\rho \epsilon_{\text{mach}}),$$

i.e.  $\|\delta A\| \leq C \rho \epsilon_{\text{mach}} \|A\|$  for small  $\epsilon_{\text{mach}}$ ,  $C$  depending on  $n$  only.

**Remark (The example again).** Swap rows:  $\tilde{P}A = \begin{bmatrix} 1 & 1 \\ \eta & 1 \end{bmatrix}$ , multiplier  $\eta$ , and  $\text{fl}(1 - \eta) = 1$  ( $\eta < \frac{1}{2} \epsilon_{\text{mach}}$ ):

$$\tilde{L}\tilde{U} = \begin{bmatrix} 1 & 0 \\ \eta & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ \eta & 1 + \eta \end{bmatrix}, \quad \frac{\|\delta A\|_{\infty}}{\|A\|_{\infty}} = \frac{\eta}{2}.$$

# How Large Can $\rho(A)$ Get?

**Proposition.**  $\rho(A) \leq 2^{n-1}$  for every nonsingular  $A \in \mathbb{C}^{n \times n}$ .

*Proof. Idea:* each step subtracts at most one entry's worth from another, so the largest entry at most doubles per step.

Let  $A^{(1)} := A$ ,  $A^{(k+1)} :=$  the matrix after step  $k$ , and  $\mu_k :=$  largest |entry| of  $A^{(k)}$ .

1. *One step at most doubles.* The swap only reorders entries. For rows  $i > k$ , using  $|l_{ik}| \leq 1$ :

$$\left| a_{ij}^{(k)} - l_{ik} a_{kj}^{(k)} \right| \leq \mu_k + \mu_k \quad \implies \quad \mu_{k+1} \leq 2\mu_k.$$

2. *Induction on  $k$ .* Base  $\mu_1 = \max |a_{jk}|$ ; then  $\mu_k \leq 2^{k-1} \mu_1$  for  $k = 1, \dots, n$ .

3. *Conclusion.*  $U = A^{(n)}$ , so  $\max |u_{jk}| = \mu_n \leq 2^{n-1} \max |a_{jk}|$ . ■

# What the Growth Bound Means

**Remark (Formally backward stable).**  $2^{n-1}$  depends on  $n$  only. So, by the theorem,

$$\frac{\|\delta A\|}{\|A\|} \leq C 2^{n-1} \epsilon_{\text{mach}} \quad \text{for all nonsingular } A \in \mathbb{C}^{n \times n} :$$

LU with partial pivoting is backward stable.

**Remark (But the constant is huge).** Some matrices reach  $\rho = 2^{n-1}$  (next lecture). For  $n = 64$ ,  $2^{63} \approx 9.2 \cdot 10^{18}$ : in double precision the guarantee says nothing useful.

**Remark (In practice).** For matrices met in practice,  $\rho(A)$  stays far below  $2^{n-1}$ , and LU with partial pivoting gives small backward error.

# $\kappa(A)$ as Stretching

**Proposition.** For nonsingular  $A$  and any lub norm,

$$\|A^{-1}\| = \frac{1}{\min_{\|z\|=1} \|Az\|}, \quad \kappa(A) = \frac{\max_{\|z\|=1} \|Az\|}{\min_{\|z\|=1} \|Az\|} :$$

largest stretch of  $A$  divided by its smallest stretch.

*Proof. Idea:*  $A^{-1}$  undoes  $A$ , so its largest stretch is  $1/(\text{smallest stretch of } A)$ .

1. The minimum exists and is  $> 0$ .  $z \mapsto \|Az\|$  is continuous on the closed, bounded set  $\|z\| = 1$ , and  $Az \neq 0$  there.

2. Substitute  $y = Az$ . As  $A$  is invertible,  $y$  runs over all nonzero vectors when  $z$  does:

$$\|A^{-1}\| = \max_{y \neq 0} \frac{\|A^{-1}y\|}{\|y\|} = \max_{z \neq 0} \frac{\|z\|}{\|Az\|} = \frac{1}{\min_{\|z\|=1} \|Az\|}.$$

3. Multiply by  $\|A\| = \max_{\|z\|=1} \|Az\|$ . ■

## $\kappa(A)$ in the 2-Norm; Distance to Singular

**Remark (2-norm).**  $\sigma_{\max} := \max_{\|z\|_2=1} \|Az\|_2$  and  $\sigma_{\min} := \min_{\|z\|_2=1} \|Az\|_2$  are the largest and smallest singular values of  $A$ :

$$\kappa_2(A) = \sigma_{\max} / \sigma_{\min}.$$

$\kappa_2(A) = 1$  exactly when  $A = cQ$ ,  $c \neq 0$ ,  $Q$  unitary: all stretches equal, the unit sphere goes to a sphere. (“If”:  $\|cQz\|_2 = |c| \|z\|_2$ . “Only if”: not shown.)

**Remark (Distance to the nearest singular matrix).** For any lub norm,

$$\min \left\{ \frac{\|\delta A\|}{\|A\|} : A + \delta A \text{ singular} \right\} = \frac{1}{\kappa(A)}.$$

“ $\geq$ ” is Step 1 of the theorem’s proof; “=” uses a  $\delta A$  as in Step 4 (not shown).  
Large  $\kappa(A) \Leftrightarrow A$  is relatively close to singular.

# Changing $b$ Instead of $A$

**Proposition.** Let  $A$  be nonsingular,  $b \neq 0$ ,  $Ax = b$  and  $A(x + \delta x) = b + \delta b$ . Then (exactly, no approximation)

$$\frac{\|\delta x\|}{\|x\|} \leq \kappa(A) \frac{\|\delta b\|}{\|b\|}.$$

*Proof.* 1. *The change.* Subtract:  $A \delta x = \delta b$ , so  $\|\delta x\| = \|A^{-1} \delta b\| \leq \|A^{-1}\| \|\delta b\|$ .  
2. *The size of  $x$ .*  $\|b\| = \|Ax\| \leq \|A\| \|x\|$ , so  $1/\|x\| \leq \|A\|/\|b\|$ .  
3. *Combine.* Multiply the two inequalities:

$$\frac{\|\delta x\|}{\|x\|} \leq \|A^{-1}\| \|\delta b\| \frac{\|A\|}{\|b\|} = \kappa(A) \frac{\|\delta b\|}{\|b\|}.$$



**Remark.** Changes in  $A$  and in  $b$  are both magnified by at most about  $\kappa(A)$ .

# Putting the LU Story Together

Everything depends on how  $\|L\| \|U\|$  compares with  $\|A\|$ .

- **No pivoting:**  $\|L\|$  can be arbitrarily large ( $2^{66}$  in the example) and  $\delta A$  as large as  $A$  itself: not stable.
- **Partial pivoting:** in the  $\infty$ -norm, using  $\max |a_{jk}| \leq \|A\|_\infty$ ,

$$\|L\|_\infty \leq n, \quad \|U\|_\infty \leq n \max |u_{jk}| = n\rho \max |a_{jk}| \leq n\rho \|A\|_\infty,$$

so  $\|L\|_\infty \|U\|_\infty \leq n^2\rho \|A\|_\infty$ : this is where  $O(\rho \epsilon_{\text{mach}})$  comes from.

**Remark (Accuracy, preview).** The two triangular solves add backward errors of the same size (proof omitted). The accuracy theorem (Lecture 14) then gives

$$\frac{\|\tilde{x} - x\|}{\|x\|} = O(\kappa(A) \rho(A) \epsilon_{\text{mach}}) :$$

problem ( $\kappa$ )  $\times$  growth ( $\rho$ )  $\times$  rounding ( $\epsilon_{\text{mach}}$ ).

# Looking Ahead

- **Next lecture:** Stability of LU with Pivoting; Interpolation